

A PERIOD-CONVENTION COUNTEREXAMPLE TO UNIVERSAL EIGENFUNCTION ASYMPTOTICS

ZHIYUAN YAO

ABSTRACT. A conjecture on phase-resonant quasiperiodic Schrödinger operators asserts exact two-sided eigenfunction tail asymptotics wherever the Lyapunov exponent exceeds the phase exponent $\delta(\alpha, \theta)$: the lower limit of $-\log\|U(n)\|/|n|$ is claimed to equal $L_V(E) - \delta(\alpha, \theta)$, for every nonconstant even real-analytic potential on $\mathbb{T} = \mathbb{R}/\mathbb{Z}$. Nothing in the statement requires 1 to be the shortest period of the potential, and we show that it is false as a consequence. The phase exponent is a purely arithmetic quantity, depending on the frequency and phase alone and not on the potential, so it cannot adapt when the potential has a smaller period. Taking $V(t) = 2e^3 \cos(4\pi t)$, of primitive period $1/2$, we construct for every standard Diophantine α a phase whose exponent in the period-one coordinate is 1 while the same operator in its primitive almost-Mathieu coordinate has exponent 2. The Lyapunov exponent is 3, so localization holds and the eigenfunctions are the objects the statement describes; but their decay is fixed in the primitive coordinate, where the sharp almost Mathieu asymptotics give the positive-tail value $3 - 2 = 1$ in place of the $3 - 1 = 2$ that the statement requires. Defining the phase exponent through the minimal fundamental period removes the counterexample, and the corrected statement is not addressed here.

1. INTRODUCTION

Let $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, let $V : \mathbb{T} \rightarrow \mathbb{R}$ be nonconstant, even and real-analytic, and consider the one-frequency quasiperiodic Schrödinger operator $H_{V,\alpha,\theta}$ with Lyapunov exponent $L_V(E)$ and phase exponent

$$\delta(\alpha, \theta) = \limsup_{|k| \rightarrow \infty} -\frac{\log\|2\theta + k\alpha\|_{\mathbb{T}}}{|k|}. \quad (1.1)$$

For an ℓ^2 eigenfunction ϕ write $U(n) = (\phi(n), \phi(n-1))^T$. The following has been conjectured for this class, as the generalization to arbitrary even analytic V of a theorem proved for the almost Mathieu operator [21, 19]:

(S) in the regime $0 < \delta(\alpha, \theta) < L_V(E)$, every ℓ^2 eigenfunction satisfies

$$\limsup_{|n| \rightarrow \infty} \frac{-\log\|U(n)\|}{|n|} = L_V(E), \quad \liminf_{|n| \rightarrow \infty} \frac{-\log\|U(n)\|}{|n|} = L_V(E) - \delta(\alpha, \theta).$$

This paper shows that (S) is false as written, and that the reason is a hypothesis it does not impose. The upper limit is not at issue; what fails is the value assigned to the lower one, and it fails because the exponent subtracted is computed against the wrong period.

The defect. Observe what (1.1) does and does not depend on. It is a statement about how well -2θ is approximated by the orbit $k\alpha$ in $\mathbb{R}/\mathbb{Z}$: an arithmetic quantity built from α and θ alone, in which the potential never appears. Statement (S) pairs it with $L_V(E)$, which does depend on V . That pairing is exactly right when 1 is the minimal period of V , which is the case for the almost Mathieu potential the statement generalizes. It is not right otherwise, because $\mathbb{R}/\mathbb{Z}$ admits perfectly good nonconstant even real-analytic functions of smaller period, and for those the operator is the same operator written in the wrong coordinate.

Take

$$V(t) = 2\lambda \cos(4\pi t), \quad (1.2)$$

which is nonconstant, even and real-analytic on $\mathbb{T}$ and so satisfies every hypothesis of (S), and whose primitive period is $1/2$. Then $V(\theta + n\alpha) = 2\lambda \cos(2\pi(2\theta + 2n\alpha))$, so $H_{V,\alpha,\theta}$ is an almost Mathieu operator, with coupling λ , frequency 2α and phase 2θ . Its phase resonance is governed by the arithmetic of that primitive pair, while (1.1) measures the arithmetic of (α, θ) . The two need not agree, and Section 3 constructs, for every standard Diophantine α , a phase at which they differ by exactly a factor of two: the stated exponent is 1, the primitive one is 2.

The consequence. Fix $\lambda = e^3$ in (1.2). The Lyapunov exponent is then 3, which exceeds the primitive exponent 2, so the operator is Anderson localized: eigenvalues and eigenfunctions exist and are exactly the objects (S) describes, and with $\delta(\alpha, \theta) = 1$ the triple lies in the regime $0 < \delta < L_V$ that (S) governs. Its eigenfunctions, however, decay according to the primitive coordinate, where the sharp almost Mathieu asymptotics fix the positive-tail lower limit at $\log \lambda - \delta(\alpha', \theta') = 3 - 2 = 1$. Statement (S) requires $L_V(E) - \delta(\alpha, \theta) = 3 - 1 = 2$. The positive-index subsequence alone already contradicts it, and the standard Diophantine frequencies have full measure.

What this is and is not. It is not a claim that the conjecture is wrong in substance. It is stated as the generalization of a theorem that is true for the almost Mathieu operator, and the counterexample exploits the one case the generalization did not intend: a potential whose stated period is not its shortest. Requiring 1 to be the minimal period of V — or, equivalently, defining the phase exponent through the minimal fundamental period — removes the counterexample at a stroke. What the paper establishes is that the hypothesis is necessary and that it is currently missing, not that the intended theorem fails. Proving the corrected statement is a separate problem and is not attempted here.

A companion statement in the same family, asserting pure point spectrum with exponentially decaying eigenfunctions in the same regime, fails to the same construction at a smaller coupling, where the operator lands on the other side of the transition; that is the subject of a separate note and nothing below depends on it.

The setting is the global theory of one-frequency quasiperiodic operators [3, 8, 6, 5, 4], resting on the dichotomy between reducibility and nonuniform hyperbolicity [7], whose arithmetic transitions descend from the Aubry–André duality and the early work of Herman, Sarnak and others [1, 17, 26, 13]. For localization in this class, which is the regime the construction below lands in, see [24, 23, 22, 9, 11, 10, 20, 2, 25, 14]; for the opposite mechanism, in which fast approximation by periodic potentials rules out eigenvalues altogether, see Gordon [16] and the singular continuous half of the almost Mathieu picture [15, 18]. See [12, 27] for surveys.

Section 2 fixes the notation and shows that one operator is described by two arithmetic pairs. Section 3 builds the phase at which the two disagree, and Section 4 derives the failure of (S). Section 5 records the hypothesis that removes it.

2. THE PERIOD CONVENTION AND THE TWO PHASE EXPONENTS

Throughout, $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ and $\|y\|_{\mathbb{T}} = \inf_{j \in \mathbb{Z}} |y - j|$. A frequency α is *standard Diophantine* when there are $\gamma > 0$ and $\tau \geq 1$ with

$$\|h\alpha\|_{\mathbb{T}} \geq \gamma |h|^{-\tau} \quad (h \in \mathbb{Z} \setminus \{0\}). \quad (2.1)$$

For $\tau > 1$ the set of α satisfying (2.1) for some $\gamma > 0$ has full Lebesgue measure: the set of α with $\|h\alpha\|_{\mathbb{T}} < |h|^{-\tau}$ has measure $O(|h|^{-\tau})$, the sum over $h \neq 0$ converges, and Borel–Cantelli leaves almost every α in only finitely many of these sets, which are then absorbed into $\gamma(\alpha)$.

Write V_λ for the potential (1.2) at coupling $\lambda > 0$; it is nonconstant, even and real-analytic on $\mathbb{T}$, and therefore admissible in (S) as it is stated. Its shortest period is $1/2$, and that single fact is the whole of the defect. Writing

$$\alpha' = 2\alpha \bmod 1, \quad \theta' = 2\theta \bmod 1, \quad (2.2)$$

we have, for every lattice site n ,

$$V_\lambda(\theta + n\alpha) = 2\lambda \cos(4\pi(\theta + n\alpha)) = 2\lambda \cos(2\pi(\theta' + n\alpha')), \quad (2.3)$$

so $H_{V_\lambda, \alpha, \theta}$ is, on the same coordinates of $\ell^2(\mathbb{Z})$ and with the same eigenfunctions, the almost Mathieu operator $H_{\lambda, \alpha', \theta'}^{\text{AMO}}$ of coupling λ .

Two arithmetic pairs therefore describe one operator: the pair (α, θ) named by (S), and the primitive pair (α', θ') that governs the operator's phase resonance. Since δ of (1.1) is built from its two arguments alone and never from the potential, nothing forces the two values to agree. Section 3 shows they need not: for every standard Diophantine α there is a phase at which

$$\delta(\alpha, \theta) = 1 \quad \text{while} \quad \delta(\alpha', \theta') = 2.$$

3. A PHASE WITH TWO PRESCRIBED EXPONENTS

The construction below is the whole of the counterexample. Prescribing *both* exponents simultaneously is what the argument needs; producing a single Liouville-type phase would not separate them.

Fix a standard Diophantine α with constants γ, τ as in (2.1). For a colour $a \in \{0, 1\}$ and a signed index $m \in \mathbb{Z}$ put

$$z_{a,m} = \frac{a}{2} - m\alpha \bmod 1, \quad \kappa_0 = 1, \quad \kappa_1 = 2. \quad (3.1)$$

Colour zero carries the integer target and colour one the half-integer target; the exponents κ_a are the two orders to be prescribed.

Lemma 3.1 (Separation of coloured centres). *Let q be a denominator of a continued-fraction convergent of α and let $|n - m| < q$. If the two centres $z_{a,n}$ and $z_{b,m}$ are distinct, then*

$$\|z_{a,n} - z_{b,m}\|_{\mathbb{T}} \geq s_q := A q^{-\tau}, \quad A = \min\{\frac{1}{2}, \gamma 2^{-\tau-1}\} > 0. \quad (3.2)$$

Proof. For equal colours and $0 < |n - m| < q$ the best-approximation property of continued fractions gives $\|z_{a,n} - z_{a,m}\|_{\mathbb{T}} = \|(n - m)\alpha\|_{\mathbb{T}} > 1/(2q)$. For different colours write $h = n - m$. If $h \neq 0$ then

$$\|z_{0,n} - z_{1,m}\|_{\mathbb{T}} = \|h\alpha - \frac{1}{2}\|_{\mathbb{T}} \geq \frac{1}{2}\|2h\alpha\|_{\mathbb{T}} \geq \gamma 2^{-\tau-1}|h|^{-\tau},$$

the middle step because $\|2y\|_{\mathbb{T}} \leq 2\|y - \frac{1}{2}\|_{\mathbb{T}}$, and the last by (2.1). If $h = 0$ the two centres are $1/2$ apart. Since $|h| < q$, each bound is at least $Aq^{-\tau}$. $\square$

Fix once and for all

$$B = 32, \quad c = 10^{-6}, \quad (3.3)$$

and for a positive index n of colour a define the compact annulus

$$\mathcal{A}_a(n) = \{y \in \mathbb{T} : c e^{-\kappa_a n} \leq \|y - z_{a,n}\|_{\mathbb{T}} \leq e^{-\kappa_a n}\}. \quad (3.4)$$

The outer radius prescribes an approximation of order κ_a at index n ; the inner radius prevents the order from being exceeded, which is what makes the resulting exponents exact rather than merely bounded below.

Lemma 3.2 (Child selection). *Let Q be a sufficiently large convergent denominator of α , let*

$$\left\lceil \frac{Q}{B} \right\rceil \leq \ell < \left\lfloor \frac{2Q}{B} \right\rfloor,$$

let $\mathcal{A}_b(\ell)$ be the current parent, and put $M = \lfloor Q/2 \rfloor$. Then for either colour a and every sufficiently large later convergent denominator q , chosen also with $q/B > M$, there is an index

$$\left\lceil \frac{q}{B} \right\rceil \leq n < \left\lfloor \frac{2q}{B} \right\rfloor \quad (3.5)$$

with

$$\mathcal{A}_a(n) \subset \mathcal{A}_b(\ell) \quad (3.6)$$

and such that every $y \in \mathcal{A}_a(n)$ satisfies

$$\|y - z_{u,m}\|_{\mathbb{T}} \geq c e^{-\kappa_u |m|} \quad (u \in \{0, 1\}, M \leq |m| < \lfloor q/2 \rfloor). \quad (3.7)$$

For the selected pair $(u, m) = (a, n)$ itself, (3.7) is the inner inequality of (3.4), so the child's own centre is not an obstruction.

Proof. Write $r = e^{-\kappa_b \ell}$ for the parent's outer radius and trim the parent to

$$\mathcal{U} = \{y : 2cr < \|y - z_{b,\ell}\|_{\mathbb{T}} < (1 - c)r\}, \quad (3.8)$$

a set of length $2(1 - 3c)r$. Irrational rotations are uniquely ergodic, uniformly in the initial point of the orbit block, so the block of consecutive candidate indices (3.5) contains at least

$$\frac{(1 - 3c)r q}{B} \quad (3.9)$$

centres lying in $\mathcal{U}$ once q is large enough. Enlarging q further makes the largest child radius at most $e^{-q/B} < cr$, so a child centred in $\mathcal{U}$ satisfies (3.6).

Call a candidate *bad* when its child meets one of the obstacle balls $B(z_{u,m}, c e^{-\kappa_u |m|})$ of (3.7) other than its own inner ball. Candidate indices obey $n < 2q/B$ and obstacle indices $|m| < q/2$, so

$$0 < |n - m| < \frac{2q}{B} + \frac{q}{2} < q \quad (3.10)$$

unless the two indices coincide, and the coinciding case with opposite colours is covered by Lemma 3.1 as well. Choose q so that the child radius is below $s_q/4$; by Lemma 3.1 an obstacle meeting the child must then have radius at least $3s_q/4$, and since obstacle radii decay exponentially in $|m|$ only $O(\log q)$ signed obstacles can do so.

Candidate centres of the fixed colour are more than $1/(2q)$ apart, so a single obstacle of radius ρ spoils at most $4q(\rho + e^{-q/B}) + 1$ candidates. Summing over both colours and both signs,

$$\#\{\text{bad candidates}\} \leq C_0 q e^{-M} + O(q e^{-q/B} \log q + \log q), \quad (3.11)$$

with C_0 independent of Q and q . Because $\ell < 2Q/B$, $\kappa_b \leq 2$ and $M = \lfloor Q/2 \rfloor$,

$$\frac{e^{-M}}{r} \leq \exp\left(-\frac{Q}{2} + \frac{4Q}{B} + O(1)\right) = e^{-3Q/8 + O(1)}. \quad (3.12)$$

Taking the initial parent scale large enough makes the first term of (3.11) smaller than half the count (3.9); for that fixed parent the remaining terms of (3.11) are $o(rq)$ as $q \rightarrow \infty$. Hence some candidate is not bad, and it is the required child. $\square$

Proposition 3.3 (Two prescribed exact orders). *Let α satisfy (2.1). There is $x \in \mathbb{T}$ with*

$$\limsup_{|m| \rightarrow \infty} \frac{-\log \|x + m\alpha\|_{\mathbb{T}}}{|m|} = 1 \quad (3.13)$$

and

$$\limsup_{|m| \rightarrow \infty} \frac{-\log \|x + m\alpha - \frac{1}{2}\|_{\mathbb{T}}}{|m|} = 2, \quad (3.14)$$

both orders being attained along subsequences of positive indices.

Proof. Begin with a sufficiently large annulus of colour zero and apply Lemma 3.2 repeatedly, alternating the requested colours $0, 1, 0, 1, \dots$ and obtaining convergent denominators $Q_1 < Q_2 < \dots$. This produces nested nonempty compact annuli, so their intersection contains a point x ; put $M_t = \lfloor Q_t/2 \rfloor \rightarrow \infty$.

The protected windows $M_t \leq |m| < M_{t+1}$ of (3.7) cover every sufficiently large signed integer, so

$$\|x - z_{a,m}\|_{\mathbb{T}} \geq c e^{-\kappa_a |m|} \quad (a = 0, 1) \quad (3.15)$$

for all large $|m|$, which after applying $-|m|^{-1} \log$ bounds both upper limits from above by κ_a , the constant c disappearing in the limit. Membership in the infinitely many selected annuli of each colour gives the matching $\|x - z_{a,n_t}\|_{\mathbb{T}} \leq e^{-\kappa_a n_t}$ along subsequences of positive indices, hence the lower bounds. With $z_{0,m} = -m\alpha$ and $z_{1,m} = \frac{1}{2} - m\alpha$ these are exactly (3.13) and (3.14). $\square$

Corollary 3.4. *Let α satisfy (2.1) and let θ satisfy $2\theta = x \bmod 1$ for the x of Proposition 3.3. Then, with α', θ' as in (2.2),*

$$\delta(\alpha, \theta) = 1, \quad \delta(\alpha', \theta') = 2, \quad (3.16)$$

and α' is again standard Diophantine.

Proof. Since $2\theta = x$, (1.1) and (3.13) give $\delta(\alpha, \theta) = 1$. For the primitive pair use the exact circle identity

$$\|2y\|_{\mathbb{T}} = 2 \min\{\|y\|_{\mathbb{T}}, \|y - \frac{1}{2}\|_{\mathbb{T}}\}, \quad (3.17)$$

applied at $y = x + k\alpha$. Because $2\theta' + k\alpha' = 2(x + k\alpha)$ modulo 1,

$$\delta(\alpha', \theta') = \max \left\{ \limsup_{k \rightarrow +\infty} \frac{-\log \|x + k\alpha\|_{\mathbb{T}}}{k}, \limsup_{k \rightarrow +\infty} \frac{-\log \|x + k\alpha - \frac{1}{2}\|_{\mathbb{T}}}{k} \right\} = 2,$$

the additive $-\log 2/k$ from (3.17) vanishing in the limit and the two upper limits being 1 and 2 by Proposition 3.3. Finally $\|k\alpha'\|_{\mathbb{T}} = \|2k\alpha\|_{\mathbb{T}} \geq \gamma(2|k|)^{-\tau}$, so α' satisfies (2.1) with constant $\gamma 2^{-\tau}$. $\square$

Remark 3.5. The two exponents are the same under either index convention. Sources differ on whether the upper limit in (1.1) is taken over $k \rightarrow +\infty$ or over $|k| \rightarrow \infty$, and in a paper about a convention that difference cannot be left implicit. Proposition 3.3 settles it: the lower bounds come from subsequences of positive indices, so they hold for the one-sided reading, while (3.15) bounds the quotient above for every sufficiently large $|m|$ of either sign, so the two-sided upper limit is no larger. Both readings therefore return $\delta(\alpha, \theta) = 1$ and $\delta(\alpha', \theta') = 2$, and nothing below depends on the choice.

Proposition 3.6 (The Lyapunov exponent is carried across). *For every $\lambda > 0$ and every energy E ,*

$$L_{V_{\lambda}, \alpha}(E) = L_{\text{AMO}, \alpha'}(E), \quad (3.18)$$

where the right-hand side is the Lyapunov exponent of the almost Mathieu cocycle of coupling λ and frequency α' .

Proof. By (2.3) the two transfer matrices agree after doubling the base point: if $A_E^{V_{\lambda}, \alpha}(N, t)$ and $A_E^{\text{AMO}, \alpha'}(N, s)$ denote the N -step matrices, then $A_E^{V_{\lambda}, \alpha}(N, t) = A_E^{\text{AMO}, \alpha'}(N, 2t)$ for every t . The doubling map $t \mapsto 2t$ preserves Haar measure on $\mathbb{T}$, so the two integrals of $N^{-1} \log \|A_E(N, \cdot)\|$ coincide for every N , and so do their limits. $\square$

4. THE WRONG TAIL EXPONENT AT THE LARGER COUPLING

Take $\lambda = e^3$ in (1.2) and let α, θ be as in Corollary 3.4. We use the sharp almost Mathieu results of Jitomirskaya and Liu [21], whose normalization is fixed by the transfer matrix

$$A(t) = \begin{pmatrix} E - 2\lambda \cos(2\pi t) & -1 \\ 1 & 0 \end{pmatrix}, \quad (4.1)$$

the convention in which (2.3) was written. At this coupling the operator sits on the localized side of the transition, so the failure is quantitative rather than spectral: the eigenfunctions (S) describes do exist, and it is the rate assigned to them that is wrong.

Theorem 4.1. *Statement (S) is false. For $V_{e^3}(t) = 2e^3 \cos(4\pi t)$ and every standard Diophantine α there is a phase θ for which $H_{V,\alpha,\theta}$ is Anderson localized in the regime (S) describes, and every ℓ^2 eigenfunction satisfies*

$$\liminf_{|n| \rightarrow \infty} \frac{-\log \|U(n)\|}{|n|} \leq 1,$$

whereas (S) asserts the value $L_{V,\alpha}(E) - \delta(\alpha, \theta) = 2$.

Proof. Fix θ as in Corollary 3.4, so that $\delta(\alpha, \theta) = 1$ and $\delta(\alpha', \theta') = 2$. By (2.3) the operator is $H_{e^3, \alpha', \theta'}^{\text{AMO}}$, and

$$\log \lambda = 3 > 2 = \delta(\alpha', \theta'), \quad (4.2)$$

so Theorem 1.1 of [21] gives Anderson localization: the spectrum is pure point with exponentially decaying eigenfunctions, and it is nonempty, so eigenvalues and eigenfunctions exist. By Proposition 3.6 and the almost Mathieu value $L = \max\{\log \lambda, 0\}$, which holds at every energy of the spectrum [11, 12] and is bounded below already by Herman's subharmonicity estimate [17], $L_{V,\alpha}(E) = \log \lambda = 3$ at every spectral energy; with $\delta(\alpha, \theta) = 1$ the pair (V, α, θ) lies in the regime $0 < \delta < L_V$ that (S) governs.

The eigenfunction asymptotics are fixed in the primitive coordinate. Corollary 1.6(ii) of [21], under the hypotheses of its Theorem 1.2, gives for every eigenfunction

$$\liminf_{n \rightarrow +\infty} \frac{-\log \|U(n)\|}{n} = \log \lambda - \delta(\alpha', \theta') = 3 - 2 = 1, \quad (4.3)$$

and eigenfunctions are not transformed by (2.3), so the two-sided lower limit is at most 1. Statement (S) requires instead

$$\liminf_{|n| \rightarrow \infty} \frac{-\log \|U(n)\|}{|n|} = L_{V,\alpha}(E) - \delta(\alpha, \theta) = 3 - 1 = 2. \quad (4.4)$$

Equations (4.3) and (4.4) are incompatible, and the positive-index subsequence in (4.3) already suffices. The standard Diophantine frequencies have full Lebesgue measure, so the frequencies exhibited here cannot be absorbed into the null set that an “almost every α ” qualifier would permit. $\square$

5. THE CORRECTED STATEMENT

Remark 5.1. Either of the following restores the intended content of (S). Require 1 to be the shortest period of V ; then $(\alpha, \theta) = (\alpha', \theta')$ in the notation of (2.2) and the two exponents coincide by construction. Alternatively, replace δ in (1.1) by a resonance exponent invariant under the finite covering reparametrizations $t \mapsto kt$ and computed over every symmetry centre of V , which for V_λ returns the value 2 that governs the operator.

The repaired statement is not addressed here, and nothing above bears on whether it is true. What Theorem 4.1 establishes is that the hypothesis is necessary and that it is currently absent: the almost Mathieu theorem of [21, 19] that (S) generalizes is stated for a potential whose shortest period is 1, and the generalization to arbitrary even analytic V carried δ across without carrying that normalization with it.

ACKNOWLEDGMENTS

Z. Y. acknowledges support from the Strategic Priority Research Program of the Chinese Academy of Sciences under Grant No. XDB1680102. Z. Y. also acknowledges support by National Natural Science Foundation of China (Grant No. 12247101). We also acknowledge Gewu Intelligence Lab for providing the collaborative research environment in which this project was developed.

Large language models were used substantively in this work: OpenAI GPT-5.6 Sol in solving the problem, including the formulation of arguments and the derivation of intermediate steps, and

Anthropic Claude Opus 5 in surveying the literature, in verifying the period convention against the source statements, in checking every attribution against the cited sources, and in drafting and revising the manuscript. The author specified the problem, guided the approach to be taken, checked every argument and result they produced, and takes full responsibility for the content of this manuscript.

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LANZHOU CENTER FOR THEORETICAL PHYSICS, KEY LABORATORY OF THEORETICAL PHYSICS OF GANSU PROVINCE, KEY LABORATORY OF QUANTUM THEORY AND APPLICATIONS OF MOE, GANSU PROVINCIAL RESEARCH CENTER FOR BASIC DISCIPLINES OF QUANTUM PHYSICS, LANZHOU UNIVERSITY, LANZHOU 730000, CHINA
Email address: yaozy@lzu.edu.cn